Milestone 4: Results — Focal Loss (H1) and the Melanoma Threshold Control (H2)

Authors: Navnit Naman (230085), Kanhaiya Kumar (230062)
Base paper: Datta et al., *Soft-Attention Improves Skin Cancer Classification Performance*, iMIMIC @ MICCAI 2021
Hypotheses tested: `M3_hypothesis.md` (H1, focal loss; decision rule in its addendum) and `M3_hypothesis2_threshold.md` (H2, threshold)
Code: `train.py` (training), `run_pod.sh` (one run per GPU pod), `M4_analyze.py` (every table and verdict below)

---

1. Summary

	Verdict	One-line reason
H1 focal loss raises melanoma recall	**Not supported.** Rejected by the §4 macro-recall criterion; *inconclusive* under the addendum's paired rule	Focal finds more melanomas (25.7 vs 20.3 per seed, pooled McNemar p = 0.011), but not on every seed (seed 7 ties). Macro recall falls (0.755 → 0.684), weighted precision falls by 0.031, and accuracy falls by 11 points.
H2 a threshold alone raises baseline recall	**Supported** on all three CE seeds	Cross-fitted at specificity ≥ 0.95, CE finds 27.1, 24.4 and 23.6 of 34 melanomas (bar: 20), with weighted precision within ±0.01 and a stable threshold.
H1 vs H2	**CE + threshold beats focal loss**	At matched specificity CE finds more melanomas: 25.0 vs 19.3 per seed at spec ≥ 0.95, and 29.3 vs 25.7 at spec ≥ 0.90. CE also keeps higher melanoma AUC (0.958 vs 0.927).

Focal loss works mainly by moving the melanoma operating point. It predicts melanoma 82–152 times (CE: 29–45) and buys recall with specificity. Once that shift is matched by a threshold on the cross-entropy model, the gain disappears and focal loss is worse. These are the outcomes we listed in M3 as *partial support at best* and *the threshold explains it*.

---

2. Setup (as pre-registered, with one hardware deviation)

- **Model:** Inception-ResNet-v2 baseline **without** Soft-Attention, cut at `block8_9`, as in our M2 reproduction.
- **Training:** Adam (lr 0.01, ε 0.1), batch 16, at most 150 epochs, EarlyStopping on val_loss (patience 30), checkpoint on best val_accuracy, melanoma class weight ×5. The training set was augmented offline to 51,699 images, and the authors' lesion-grouped split was used (828 test images, 34 melanomas). The test set also serves as validation (the paper's protocol, see §6).
- **The single change:** loss = `categorical_crossentropy` vs `CategoricalFocalCrossentropy(γ = 2, α = 1)`. Everything else is identical.
- **Runs:** 2 losses × seeds 42, 7, 2024 = 6 runs, one per pod. TF 2.20, NVIDIA **A40** (RunPod secure cloud). The addendum named an RTX 4090, but none was available, so all six runs used the same A40 type. 17.6 GPU-hours of training in total.

Run	Epochs run	Best epoch	Train time (h)
CE seed 7	49	23	2.14
CE seed 42	48	22	2.14
CE seed 2024	42	36	1.81
Focal seed 7	112	67	4.69
Focal seed 42	95	65	3.97
Focal seed 2024	65	29	2.86

Focal runs train about twice as long and pick their best checkpoint much later. Its loss values are smaller, so the same absolute `min_delta` keeps resetting early stopping. This is the step-size / convergence risk from M3 §5.

---

3. Results (arg-max decision, test set n = 828)

3.1 Per run

Loss	Seed	Accuracy	W. precision	Macro recall	mel found	mel recall (95 % Wilson)	mel spec.	mel AUC	W. AUC	mel predicted
CE	7	0.924	0.930	0.808	26/34	0.765 (0.60–0.88)	0.976	0.951	0.979	45
CE	42	0.918	0.914	0.735	16/34	0.471 (0.31–0.63)	0.977	0.957	0.984	34
CE	2024	0.924	0.921	0.722	19/34	0.559 (0.39–0.71)	0.987	0.965	0.986	29
Focal	7	0.860	0.906	0.786	26/34	0.765 (0.60–0.88)	0.929	0.943	0.966	82
Focal	42	0.825	0.887	0.651	25/34	0.735 (0.57–0.85)	0.895	0.916	0.963	108
Focal	2024	0.755	0.878	0.616	26/34	0.765 (0.60–0.88)	0.841	0.923	0.952	152

3.2 Mean ± std over the three seeds

Loss	Accuracy	W. precision	Macro recall	mel recall	mel spec.	mel AUC	W. AUC
CE	**0.922 ± 0.003**	**0.922 ± 0.008**	**0.755 ± 0.046**	0.598 ± 0.151	**0.980 ± 0.006**	**0.958**	**0.983 ± 0.004**
Focal	0.813 ± 0.054	0.890 ± 0.014	0.684 ± 0.090	**0.755 ± 0.017**	0.889 ± 0.044	0.927	0.961 ± 0.008

3.3 Where focal loss's errors go (correct / total, per class)

Run	akiec	bcc	bkl	df	mel	nv	vasc
CE 7	18/23	21/26	44/66	4/6	26/34	642/663	10/10
CE 42	11/23	21/26	44/66	5/6	16/34	654/663	9/10
CE 2024	16/23	18/26	48/66	3/6	19/34	652/663	9/10
Focal 7	14/23	24/26	38/66	5/6	26/34	596/663	9/10
Focal 42	11/23	16/26	31/66	4/6	25/34	589/663	7/10
Focal 2024	14/23	6/26	33/66	3/6	26/34	534/663	9/10

Compared with CE on the same seed, focal loss gets 46–118 more *nv* images wrong and about 11 more *bkl* images, while predicting melanoma 37–123 more times. On seed 2024 it also collapses on *bcc* (6/26). The extra melanomas cost far more errors on the other classes, which is why macro recall falls.

3.4 Paired test, same seed, same 828 images (exact McNemar)

Seed	All images: only CE right / only focal right	p	Melanomas: only CE found / only focal found	p
7	75 / 22	< 0.001	1 / 1	1.000
42	104 / 27	< 0.001	5 / 14	0.064
2024	161 / 21	< 0.001	4 / 11	0.118
Pooled (102 melanomas)			**10 / 26**	**0.011**

On every seed CE is significantly more often right overall. On melanomas, focal is ahead when pooled but ties on seed 7.

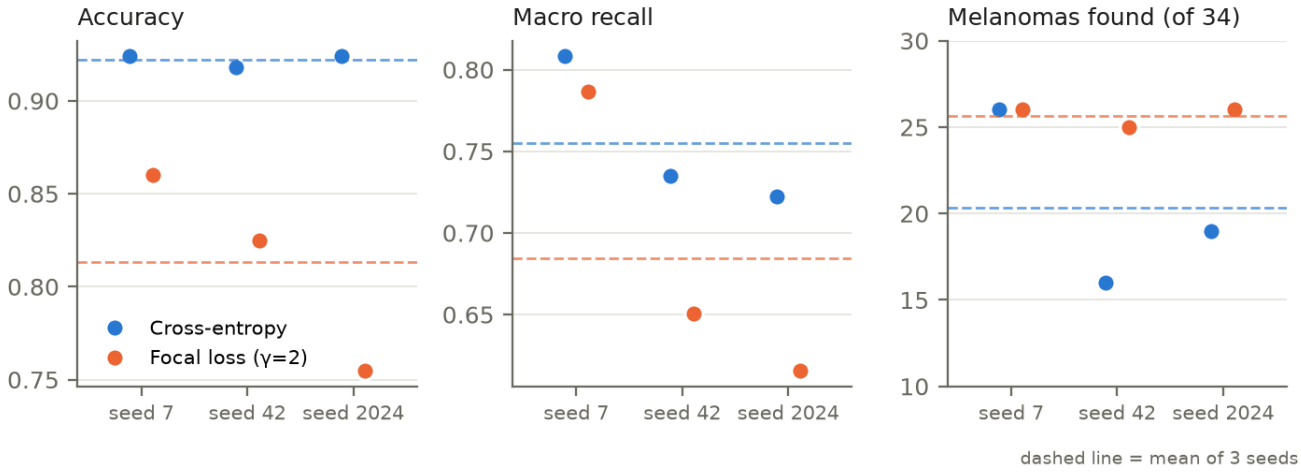


Figure 1. Each dot is one run; dashed lines are the mean over three seeds. Focal loss raises and stabilises the melanoma count but lowers accuracy and macro recall on every seed.

4. H1 verdict under the pre-registered rules

Criterion (written before any M4 run)	Result	Met?
§4: focal mean melanoma recall exceeds the CE mean by > 1 CE std	+5.3 melanomas vs CE std 5.1	yes (barely)
Addendum: focal finds more melanomas on **every** seed	26 = 26 on seed 7	**no**
Addendum: pooled melanoma McNemar $p < 0.05$	$p = 0.011$	yes
§4: macro recall improves (not only through df / vasc)	0.755 → 0.684	**no: reject**
§4 / addendum: weighted precision falls by ≤ 0.02 (otherwise *partial* at best)	−0.031	**no**

H1 is not supported. Under the addendum's paired rule alone it is *inconclusive* (this is what `M4_analyze.py` prints). The §4 macro-recall criterion, also fixed in advance, rejects it outright. Even if both of those were ignored, the precision drop would cap it at *partial support*. We do not report this as a positive result.

Run-to-run variance is itself a finding. CE melanoma recall ranges from 16 to 26 of 34 across seeds with identical code and data. So M2's single-run comparison (IRv2 15/34 vs IRv2 + SA 21/34) is within one seed's noise. Focal loss is much more stable on melanoma (25–26 of 34) but much less stable overall (accuracy 0.755–0.860).

5. H2: cross-fitted melanoma threshold

The threshold is picked on 4/5 of the test set (stratified) and applied to the remaining 1/5, repeated over 5 folds × 20 random splits. No image is scored by a threshold chosen on itself.

Model (mean of 3 seeds)	Operating point	mel found / 34	mel spec.	W. precision
CE	arg-max	20.3	0.980	0.922
CE	**spec ≥ 0.95**	**25.0**	0.950	0.921
CE	spec ≥ 0.90	**29.3**	0.899	0.923
CE	spec ≥ 0.713 (dermatologists, sens. 0.866)	33.3	0.713	0.939
Focal	arg-max	25.7	0.889	0.890
Focal	spec ≥ 0.95	19.3	0.949	0.881
Focal	spec ≥ 0.90	25.7	0.899	0.886
Focal	spec ≥ 0.713	32.8	0.712	0.895

Per seed, CE at spec ≥ 0.95 finds **27.1, 24.4 and 23.6** of 34 (threshold std across folds 0.026, 0.022, 0.013), and weighted precision moves by −0.007, +0.008 and −0.002. All three H2 criteria hold on all three seeds (≥ 20 / 34, Δ precision within 0.02, threshold std < 0.2), so **H2 is supported**. The full per-seed table is in `M4_analyze.py` output §4.

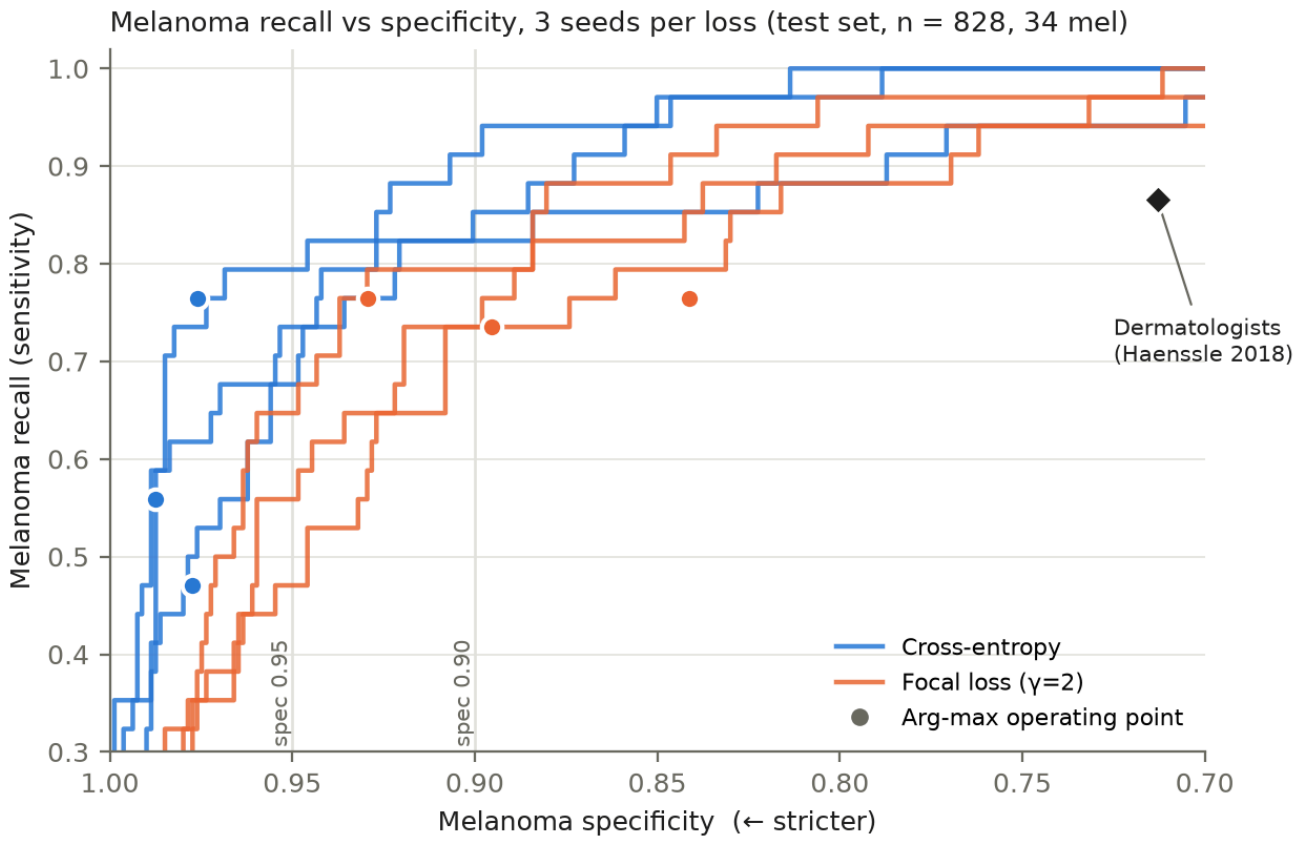


Figure 2. Melanoma recall against specificity as the melanoma threshold moves, one line per run (blue = CE, orange = focal). Dots are the default arg-max decisions. The curves are drawn on the full test set for illustration; the numbers in the table above are cross-fitted. The CE curves lie above the focal curves at almost every specificity, so focal's extra melanomas come from moving along a lower curve, not from a better one.

The key comparison. Focal loss at arg-max sits at specificity 0.889. CE thresholded to specificity 0.90 finds **29.3** melanomas at that point, against focal's **25.7**, with higher weighted precision (0.923 vs 0.890). Focal loss achieves less than a zero-cost threshold on the model we already had. This matches the M3 H2 table's *Supported* row: arg-max recall alone cannot show that a loss helps.

Two cautions for H2: - It was pre-registered on the M2 Kaggle seed-42 checkpoint (15/34), whose probabilities were not saved. It is tested here on the three new CE runs instead. - The IRv2 + Soft-Attention arm of H2 was not run in M4.

---

6. Limitations

- **Test set used as validation** (paper protocol): checkpoint selection and early stopping both see the test set. This affects CE and focal equally, but all absolute numbers are optimistic.
- **34 test melanomas.** One image is 2.9 recall points. Three seeds give a rough standard deviation, which is why the paired test was added in advance.
- **Focal loss was run with the paper's optimiser settings**, tuned for cross-entropy (lr 0.01, ε 0.1). A focal-specific learning rate or a longer schedule might change the result. That would be a second variable, so it was not done here (§7).
- **Hardware:** A40 instead of the planned RTX 4090 and M2's Kaggle T4. CE seed 42 on the A40 found 16/34, close to the M2 Kaggle run's 15/34.

- Benchmark numbers on a curated research dataset. **No clinical claims.**

---

7. What we would do next

1. **Threshold first.** Report every model at matched specificity (or by AUC) rather than at arg-max. The M2 attention result should be re-checked this way, across seeds.
2. Retest focal loss with its own learning rate (e.g. 1e-3) and early stopping on a fixed held-out validation split, as a new, separately pre-registered hypothesis.
3. Move the evaluation to a validation split that is separate from the test set, then repeat both arms.

Per-run outputs (test-set probabilities, training histories, per-class metrics, confusion matrices and curves) are in `m4_results/runs/`. *Every table above is regenerated by* `python M4_analyze.py m4_results/runs` *(full output:* `m4_results/M4_results_tables.md`*); the figures by* `make_m4_figures.py`. *Model weights are not committed.*

References. Datta S. K. et al., iMIMIC @ MICCAI 2021. Lin T.-Y. et al., *Focal Loss for Dense Object Detection*, ICCV 2017. Haenssle H. A. et al., *Annals of Oncology* 29(8), 2018.